

Figure 1 | SSTBA turns study-specific spatial atlases into auditable state and boundary outputs

Frozen signatures, explicit graph choices, falsification diagnostics and transfer without retraining



Three interpretable outputs

Mechanical-border	cardiomyocyte edge and border-zone direction
Immune-fibrotic	myeloid, TGF- β and fibro-inflammatory activity
Fibroblast-scar repair	POSTN/CTHRC1-associated reparative scar state

Required diagnostics

- 1 missing-gene and module coverage
- 2 sample omission and module dropout
- 3 matched random signatures and spatial nulls
- 4 simple-score baselines and negative results

Reproducibility contract

CLI validation • deterministic test run • versions and parameters
input/output SHA-256 hashes

14,147 mouse spots
+ independent mouse and human sections